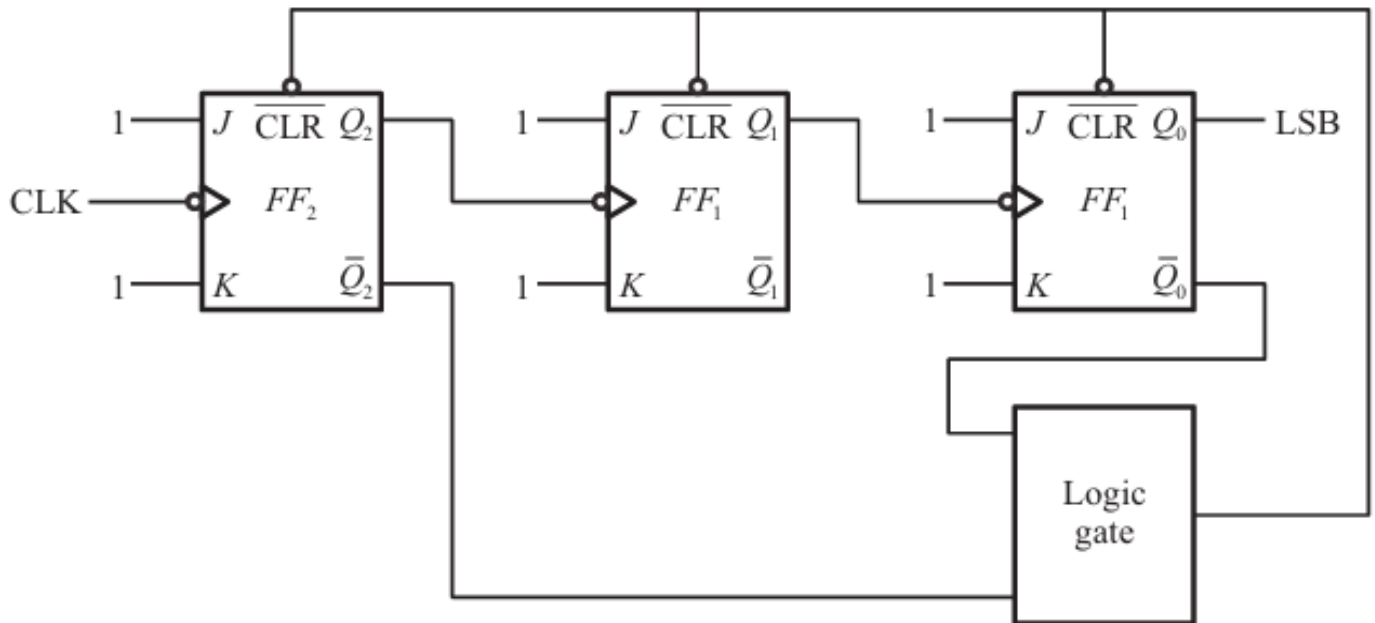


Tutorial-3 Digital Systems EE1201 -2026

1.

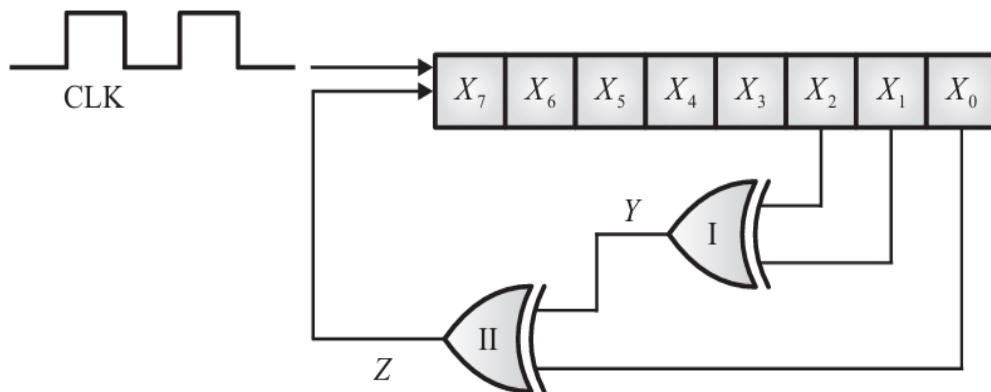
Consider the counter circuit shown in figure, with all the Flip-Flop in reset condition.



If the circuit works as MOD-5 counter, then the logic gate used is

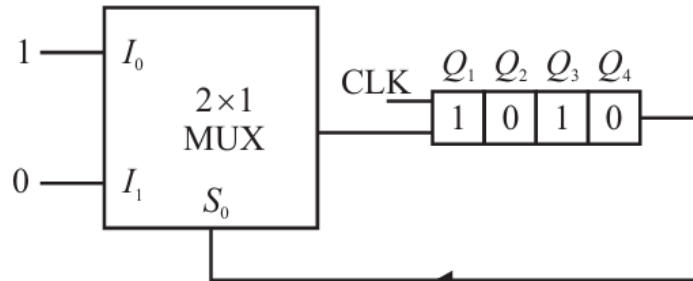
2.

The initial contents of right-shift register shown below are 01101110. The contents of the shift register after 3 clock pulses will be



3.

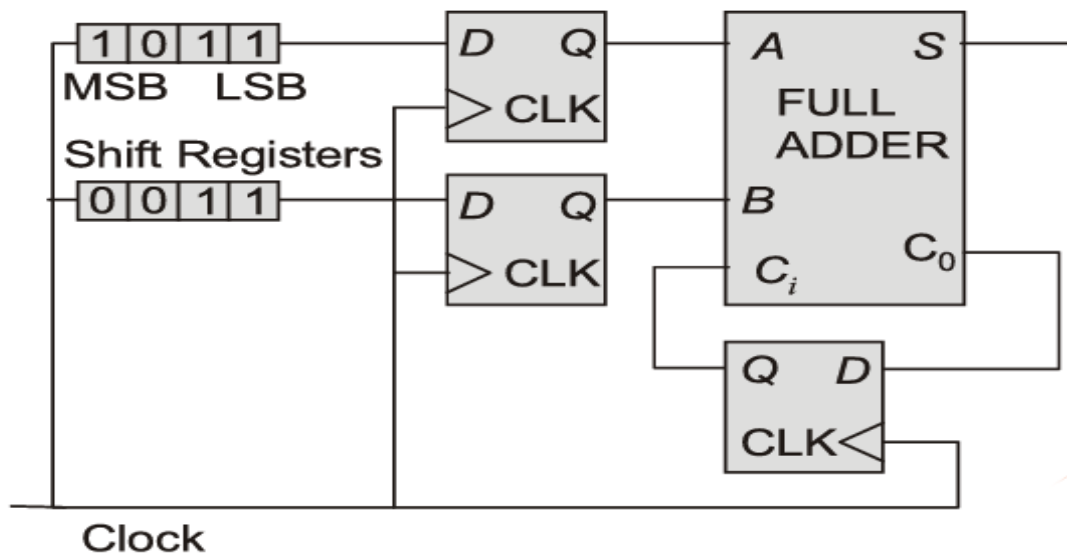
The initial content of the 4-bit right shift, parallel in parallel out shift register is 1010. Input of shift register is derived from 2×1 MUX.



The number of clock pulses after which the initial pattern of shift register reappears (in integer).

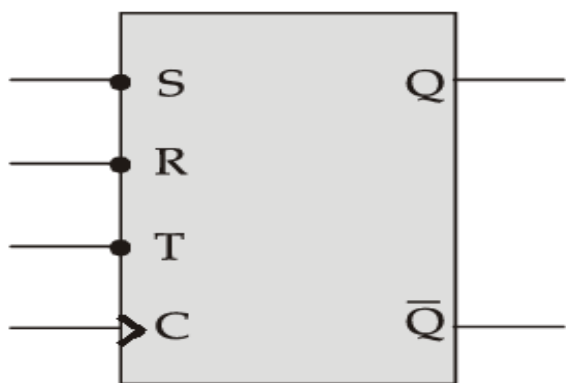
4.

For the circuit shown in the figure below, two 4-bit parallel-in serial-out shift registers loaded with the data shown are used to feed the data to a full adder. Initially, all the flip-flops are in clear state. After applying two clock pulses, the outputs of the full adder should be



5. write down the characteristic equation of the flip flop

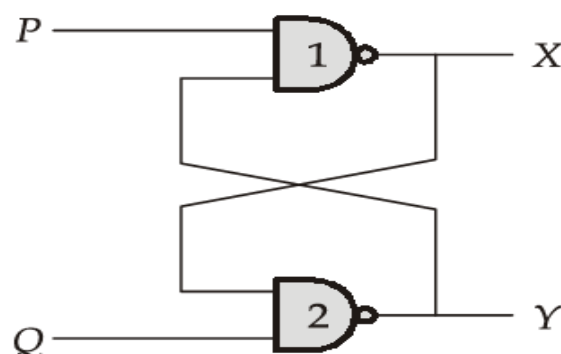
Amazon announce a new flip flop named Set-Reset-Toggle due to shortage elements to the electronics and computer industries. The device symbol and function table for this flip flop are shown below:



S	R	T	Function
X	X	1	Toggle
1	0	0	Set
0	1	0	Reset
0	0	0	Hold
1	1	0	Hold

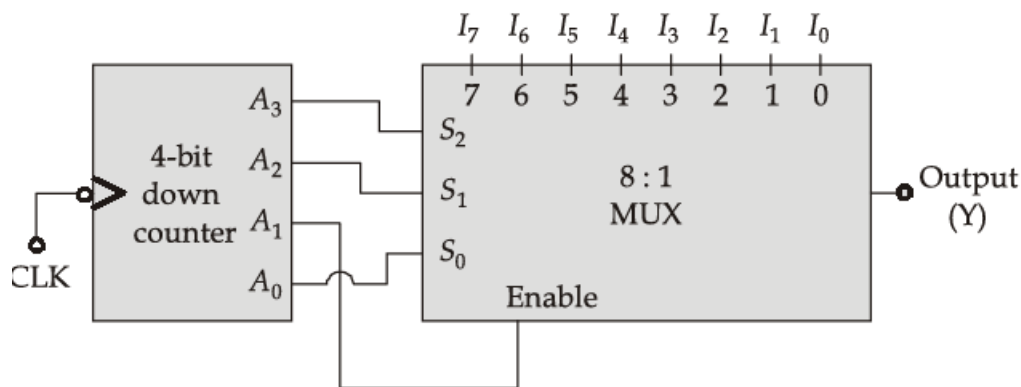
6.

In the latch circuit shown, the NAND gates have non-zero, but unequal propagation delay. The present input condition is ; $P = Q = '0'$. If the input condition is changed simultaneously to $P = Q = '1'$, the outputs X and Y are

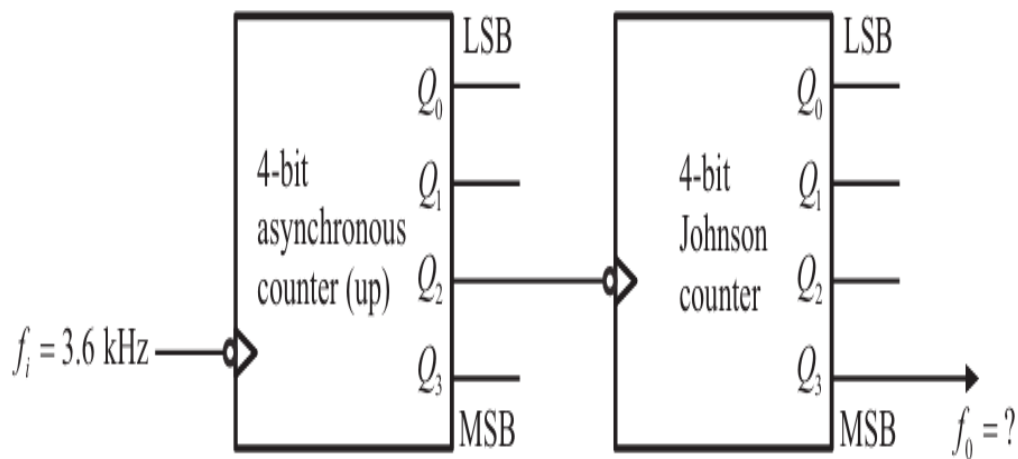


7.

A 4-bit Down counter is used to control the output of the multiplexer as shown in figure. The counter is initially at $(1111)_2$, then the output of the multiplexer will follow the sequence



8.



If the input clock frequency of up-counter is 3.6 kHz, frequency of output wave form ' f_0 ' (in Hz) is _____.

9a) An XY flip flop works as follows

If XY 00, next state is 0

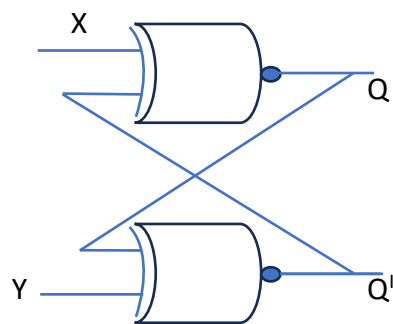
If XY 01, state remains same

If XY 10, state gets complemented

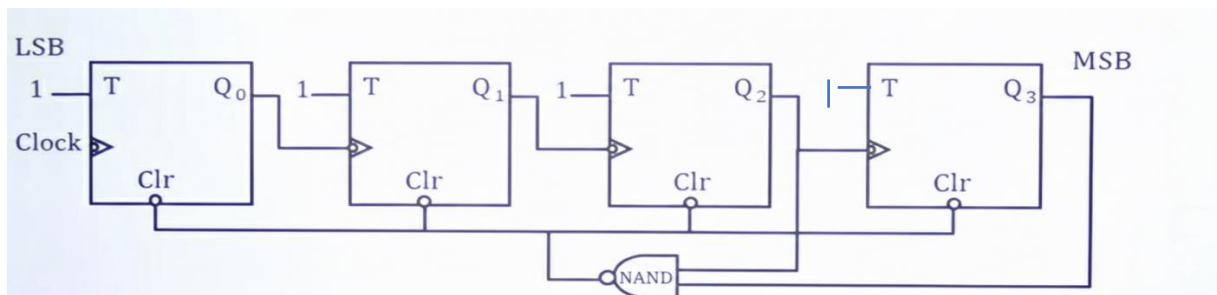
If XY 11, next state is 1

The characteristic equation is

9b) Write the characteristic table for the following.



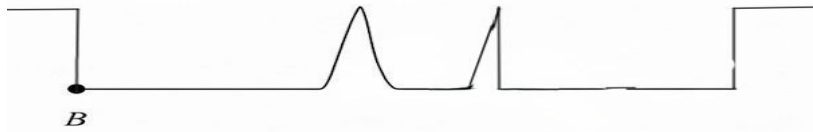
10) MOD number of the counter is



11a) The bit sequence 0010 is serially entered (right-most bit first) into a 4-bit parallel out shift register that is initially clear. What are the Q outputs after two clock pulses?

11b) A 4-bit shift register is storing the nibble 1101. The nibble 1011 is waiting to be entered on the serial data-input line. After three clock pulses, the shift register is storing.

12) If the switch is closed, the metal contacts bounce before coming to rest and input at B is as shown what will be the output waveforms for the given circuits below.



Input Waveform at Switch

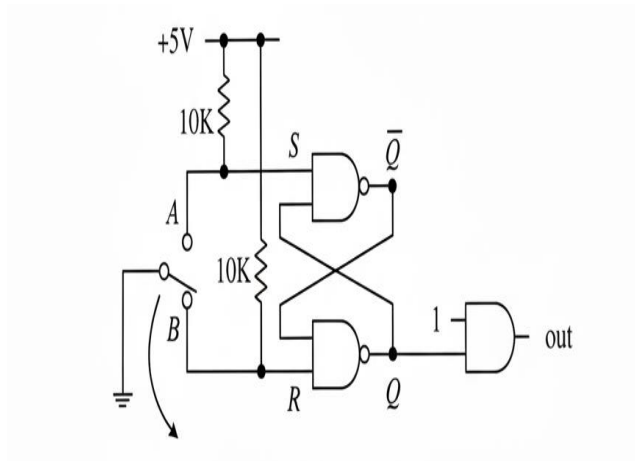


Fig a

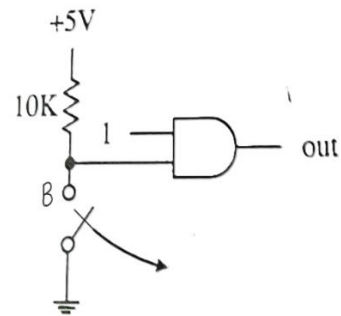
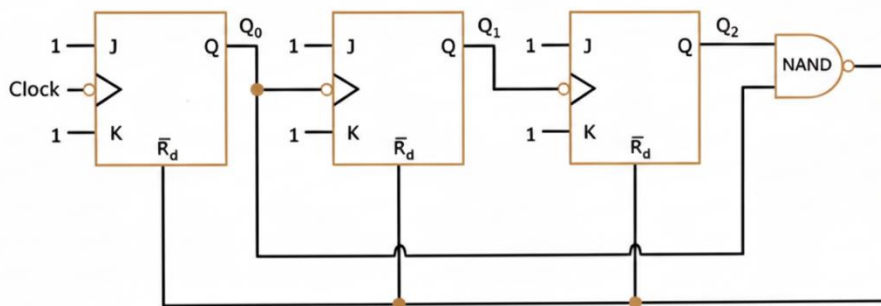


Fig b

13) The circuit shown consists of J-K flip-flops, each with an active low asynchronous reset. The MOD counter corresponding to this is



14a) Design a Synchronous Counter by using T Flip Flop which count the sequence 0-2-3-0.

14b) Design a Synchronous Counter by using T Flip Flop which count the sequence 1-2-3-5-7-1 and make it self-starting such that unused states transition to state 1.

